

EMLM_26_Group11

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2026-03-13

Case study: Longitudinal Progression of Motor Symptoms in Parkinson's Disease

Variables

- id: patient indicator.
- sex: sex indicator (1: males, 2: females).
- Age: age of the patient at inclusion.
- smoking_status: Smoking history at baseline (0: Never smoker, 1: Former smoker, 2: Current smoker).
- Visit: Visit number, i.e., years since inclusion.
- UPDRS: UPDRS Part III score at each of the 8 visits, respectively.

Question 1: Present summary statistics for the sample, including the number of males/females, age distribution at diagnosis, and distribution of smoking status.

Data Exploration

```
#Factorize sex and smoking_status, set age and UPDRS as numeric
DF <- DF %>%
  mutate(
    sex = as.factor(sex),
    smoking_status = factor(smoking_status, levels = c(0, 1, 2),
                           labels = c("Never", "Former", "Current")),
    Visit = as.integer(Visit),
    Age = as.numeric(Age),
    UPDRS = as.numeric(UPDRS)
  )

# Patient-level dataset (Age/sex/smoking_status repeat across visits in long format)
DF_pat <- DF %>%
  distinct(id, .keep_all = TRUE) %>%
  select(id, sex, Age, smoking_status)
```

Univariate Descriptive Statistics

Age

The dataset contained complete information on age at inclusion. The mean age was 64.64 years (SD = 13.49), with a median of 65.1 years. The interquartile range (IQR = 23.75) indicates that the middle 50% of patients were between 52.58 and 76.33 years old. Ages ranged from 41.1 to 88 years, and no implausible values were observed. The histogram and density curve suggest a fairly broad distribution spanning middle to older adulthood. The distribution appears approximately symmetric overall, although the Q–Q plot indicates deviations from normality in the tails, particularly at the lower and upper extremes. The boxplot does not show extreme outliers. *Overall, the age distribution reflects a typical Parkinson's disease cohort, with most patients diagnosed in their 50s to 70s and a moderate spread across older age groups.*

```
# 1 - Age at inclusion
# Mean, SD, Median, IQR, Min, Max
age_stats <- DF_pat %>%
  summarise(
    n = sum(!is.na(Age)),
    missing = sum(is.na(Age)),
    mean = mean(Age, na.rm = TRUE),
    sd = sd(Age, na.rm = TRUE),
    median = median(Age, na.rm = TRUE),
    q1 = quantile(Age, 0.25, na.rm = TRUE),
    q3 = quantile(Age, 0.75, na.rm = TRUE),
    iqr = IQR(Age, na.rm = TRUE),
    min = min(Age, na.rm = TRUE),
    max = max(Age, na.rm = TRUE)
  )
age_stats
```

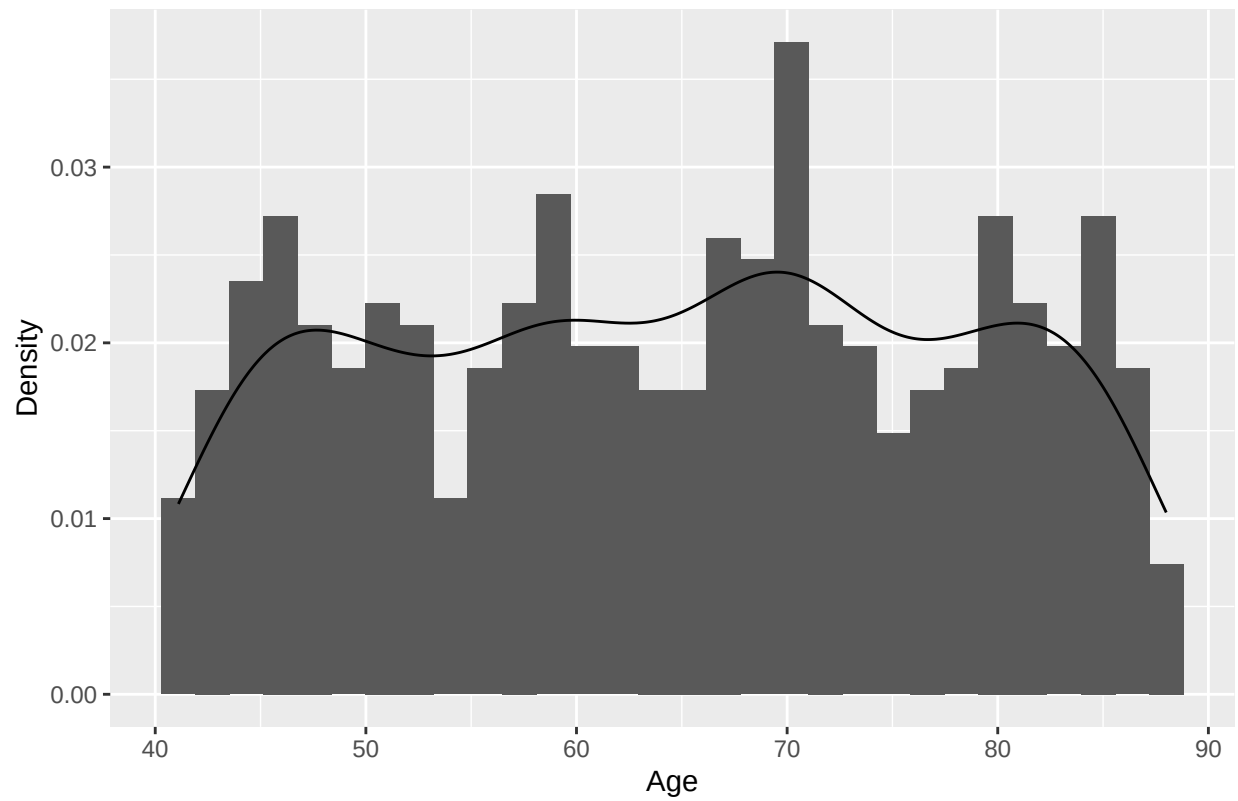
```
##      n missing   mean    sd median    q1    q3   iqr  min max
## 1 500         0 64.6414 13.4897   65.1 52.575 76.325 23.75 41.1 88
```

```
# Check for impossible values (example thresholds)
DF_pat %>%
  filter(!is.na(Age), (Age < 30 | Age > 100)) %>%
  arrange(Age)
```

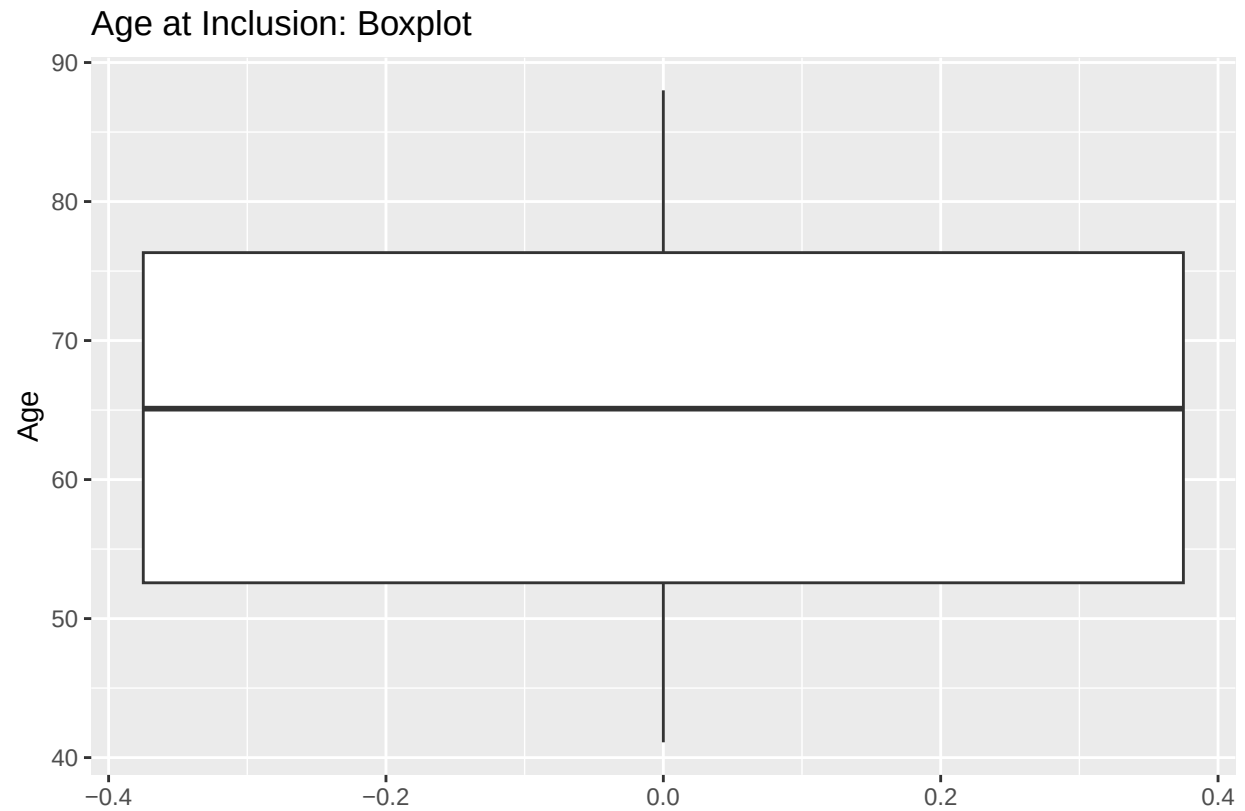
```
## [1] id          sex          Age          smoking_status
## <0 rows> (or 0-length row.names)
```

```
# Histogram + density
ggplot(DF_pat, aes(x = Age)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30, na.rm = TRUE) +
  geom_density(na.rm = TRUE) +
  labs(title = "Age at Inclusion: Histogram + Density", x = "Age", y = "Density")
```

Age at Inclusion: Histogram + Density

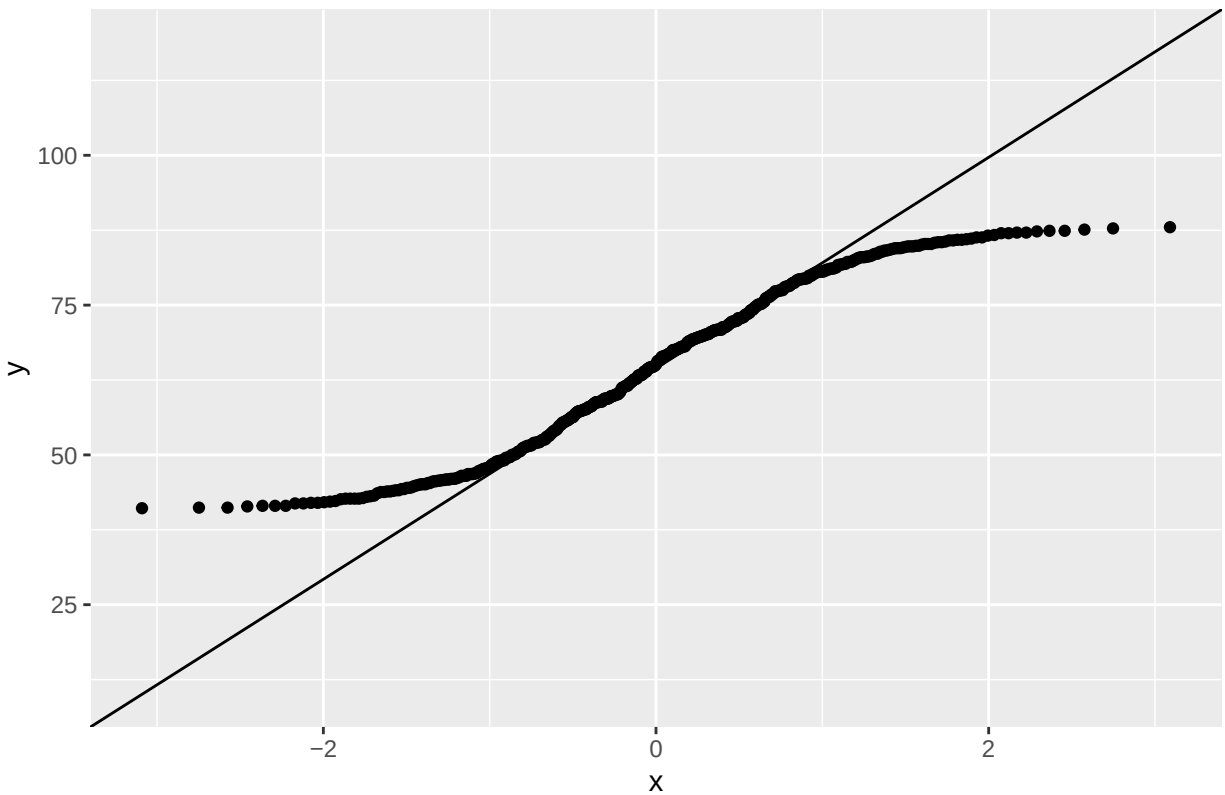


```
# Boxplot
ggplot(DF_pat, aes(y = Age)) +
  geom_boxplot(na.rm = TRUE) +
  labs(title = "Age at Inclusion: Boxplot", x = "", y = "Age")
```



```
# Optional: Q-Q plot for approximate normality check  
ggplot(DF_pat, aes(sample = Age)) +  
  stat_qq(na.rm = TRUE) +  
  stat_qq_line(na.rm = TRUE) +  
  labs(title = "Age at Inclusion: Q-Q Plot")
```

Age at Inclusion: Q-Q Plot



Sex

The dataset contains a balanced sample of male and female patients.

```
sex_tab <- DF_pat %>%
  count(sex, name = "n") %>%
  mutate(prop = n / sum(n))
sex_tab
```

```
##      sex    n prop
## 1   male  250  0.5
## 2 female  250  0.5
```

Smoking Status

The majority of patients included in the dataset have never been smokers, slightly less than one third are former smokers, and around 20% are current smokers.

```
smoke_tab <- DF_pat %>%
  count(smoking_status, name = "n") %>%
  mutate(percent = 100 * n / sum(n))
smoke_tab
```

```
##      smoking_status    n percent
## 1           Never  261    52.2
## 2          Former  141    28.2
## 3           Current   98    19.6
```

UPDRS overall (all visits pooled)

Across all 2,762 observed visits, the mean UPDRS Part III score was 26.02 (SD = 16.28), with a median of 22.73. The interquartile range (13.98–34.63) indicates that 50% of the observations fall within a moderately wide band of motor severity. Scores ranged from 0.22 to 106.94, covering nearly the full theoretical scale (0–108). No floor or ceiling effects were observed, as no observations occurred at the minimum or maximum possible values. The histogram and density curve reveal a clearly right-skewed distribution, with most observations concentrated between approximately 10 and 40 points and a long tail extending toward higher values. This is further supported by the fact that the mean exceeds the median, consistent with positive skewness. The boxplot also highlights several high-value outliers, reflecting a subset of visits with severe motor impairment. Overall, the pooled UPDRS distribution suggests that most patients were in the mild-to-moderate range of motor severity, with fewer observations at very high levels of impairment.

The values of the outliers lie within the score range consistent with severe PD (70–100) and are therefore possible in a clinical sample. Further in the exploratory data analysis, we will examine if high values cluster by visit (high UPDRS showing consistently in later visits might signal disease progression), or by patient (patient with high range throughout the study, or isolated spikes), particularly if the extreme values are already present at Visit 0.

Because the current analysis is longitudinal in nature, the resulting model will most likely assume normality of residual. In a later analysis, residual diagnostics will be performed and, if necessary, we might elect to model time non-linearly and/or use robust SEs.

```
updrs_overall_stats <- DF %>%
  summarise(
    n = sum(!is.na(UPDRS)),
    missing = sum(is.na(UPDRS)),
    mean = mean(UPDRS, na.rm = TRUE),
    sd = sd(UPDRS, na.rm = TRUE),
    median = median(UPDRS, na.rm = TRUE),
    q1 = quantile(UPDRS, 0.25, na.rm = TRUE),
    q3 = quantile(UPDRS, 0.75, na.rm = TRUE),
    min = min(UPDRS, na.rm = TRUE),
    max = max(UPDRS, na.rm = TRUE)
  )
updrs_overall_stats
```

```
##      n missing      mean      sd  median      q1      q3  min      max
## 1 2762         0 26.01858 16.28025 22.73165 13.98022 34.6272 0.22 106.9429
```

```
# Range check (should be 0-108)
DF %>%
  filter(!is.na(UPDRS), (UPDRS < 0 | UPDRS > 108)) %>%
  arrange(UPDRS)
```

```
## [1] id          UPDRS          Visit          Age          sex
## [6] smoking_status
## <0 rows> (or 0-length row.names)
```

```
# Floor/ceiling effects (overall)
floor_ceiling_overall <- DF %>%
  summarise(
    floor_n = sum(UPDRS == 0, na.rm = TRUE),
    floor_pct = 100 * mean(UPDRS == 0, na.rm = TRUE),
```

```

    ceiling_n = sum(UPDRS == 108, na.rm = TRUE),
    ceiling_pct = 100 * mean(UPDRS == 108, na.rm = TRUE)
  )
floor_ceiling_overall

```

```

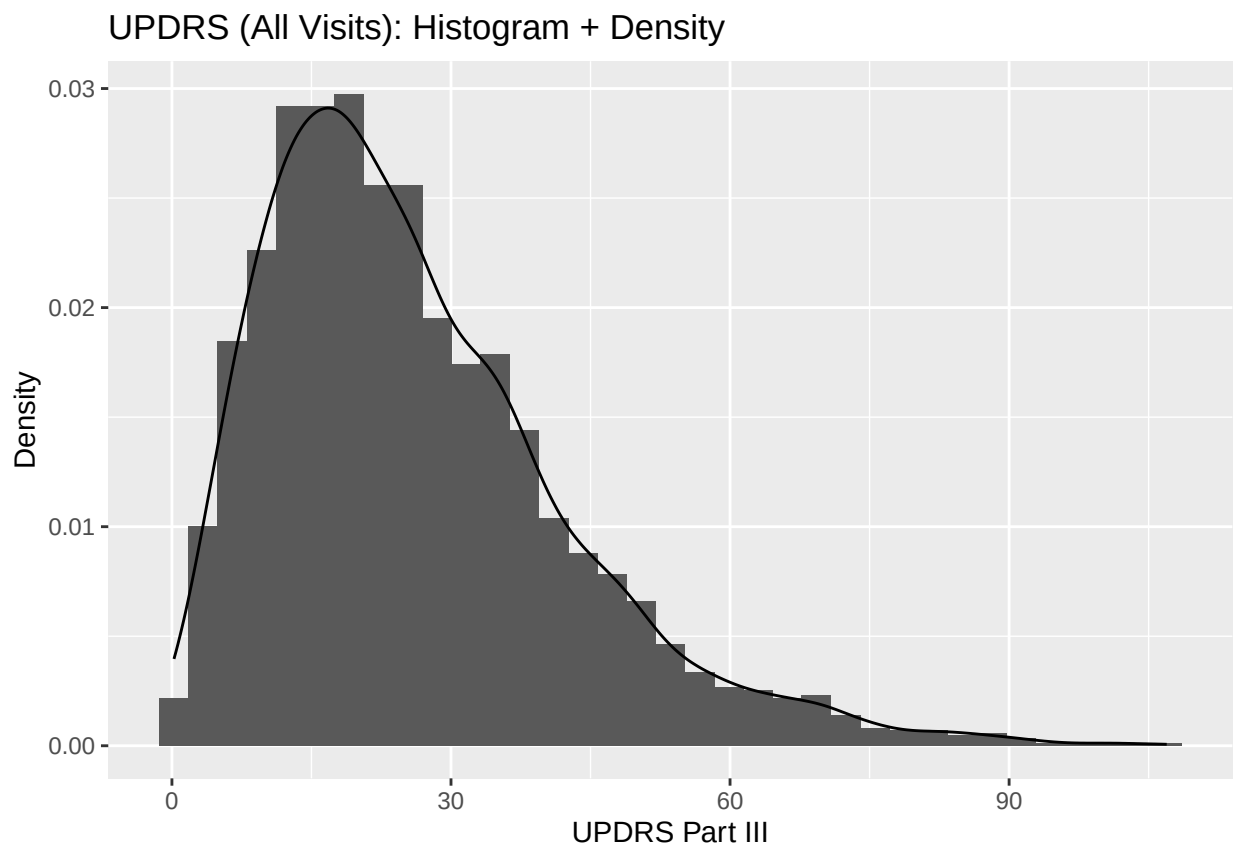
##   floor_n floor_pct ceiling_n ceiling_pct
## 1      0         0         0         0

```

```

# Overall plots: Histogram + density, boxplot
ggplot(DF, aes(x = UPDRS)) +
  geom_histogram(aes(y = after_stat(density)), bins = 35, na.rm = TRUE) +
  geom_density(na.rm = TRUE) +
  labs(title = "UPDRS (All Visits): Histogram + Density", x = "UPDRS Part III", y = "Density")

```

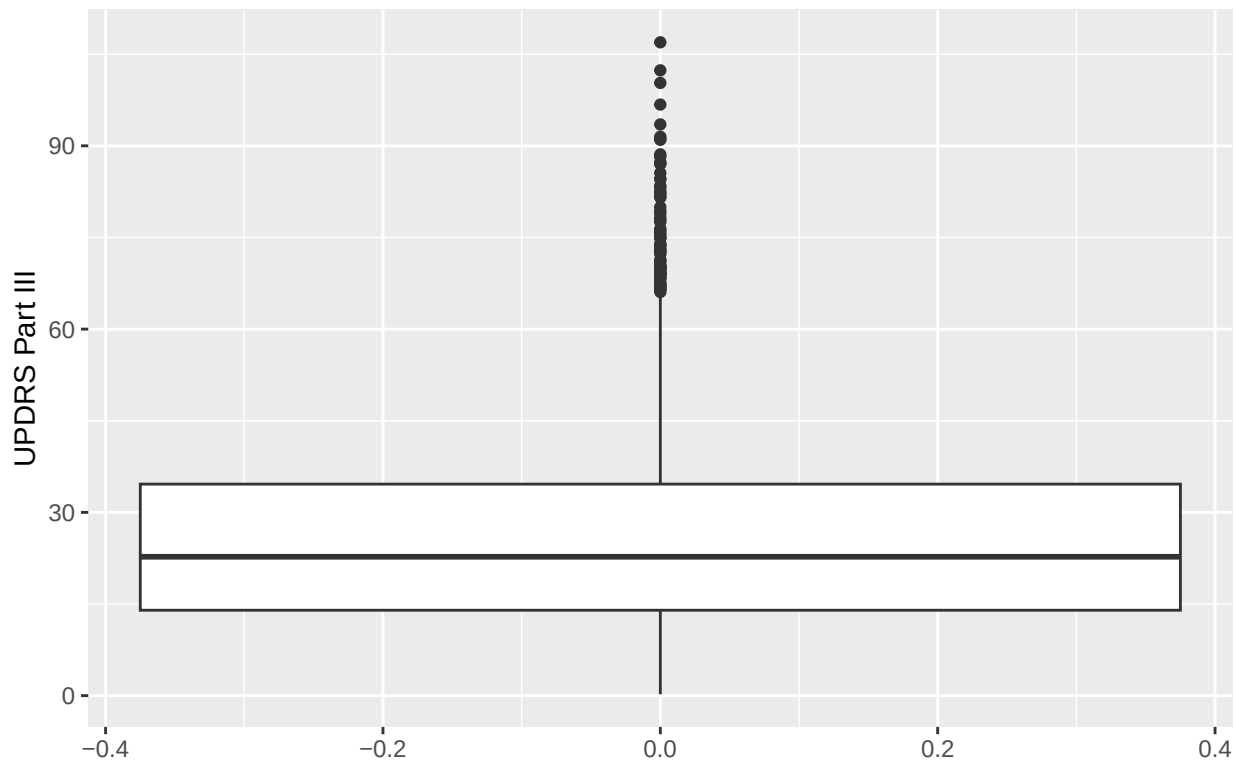


```

ggplot(DF, aes(y = UPDRS)) +
  geom_boxplot(na.rm = TRUE) +
  labs(title = "UPDRS (All Visits): Boxplot", x = "", y = "UPDRS Part III")

```

UPDRS (All Visits): Boxplot



```
# Skewness: compare mean vs median
DF %>%
  summarise(mean = mean(UPDRS, na.rm = TRUE),
            median = median(UPDRS, na.rm = TRUE))
```

```
##      mean  median
## 1 26.01858 22.73165
```

UPDSR at baseline

At baseline (Visit 0), UPDRS Part III scores were available for all 500 patients, with no missing observations. The mean baseline score was 17.24 (SD = 8.94), and the median was 16.02, indicating that most patients presented with mild to moderate motor impairment at inclusion. The interquartile range (10.39–22.91) shows that 50% of patients had scores within this relatively narrow band, suggesting moderate variability in disease severity at diagnosis. Scores ranged from 0.23 to 51.85, and no floor or ceiling effects were observed. The histogram and density plot reveal a mildly right-skewed distribution, with a longer tail toward higher scores, while the boxplot identifies a small number of higher values above the upper whisker. These higher baseline scores are clinically plausible and likely represent patients who entered the study with more advanced motor symptoms.

```
DF_base <- DF %>%
  filter(Visit == 0)

updrs_base_stats <- DF_base %>%
  summarise(
```

```

n = sum(!is.na(UPDRS)),
missing = sum(is.na(UPDRS)),
mean = mean(UPDRS, na.rm = TRUE),
sd = sd(UPDRS, na.rm = TRUE),
median = median(UPDRS, na.rm = TRUE),
q1 = quantile(UPDRS, 0.25, na.rm = TRUE),
q3 = quantile(UPDRS, 0.75, na.rm = TRUE),
min = min(UPDRS, na.rm = TRUE),
max = max(UPDRS, na.rm = TRUE)
)
updrs_base_stats

```

```

##      n missing   mean      sd  median      q1      q3   min    max
## 1 500         0 17.2394 8.944131 16.02305 10.38598 22.9129 0.2276 51.8539

```

```

# Floor/ceiling at baseline
floor_ceiling_base <- DF_base %>%
  summarise(
    floor_n = sum(UPDRS == 0, na.rm = TRUE),
    floor_pct = 100 * mean(UPDRS == 0, na.rm = TRUE),
    ceiling_n = sum(UPDRS == 108, na.rm = TRUE),
    ceiling_pct = 100 * mean(UPDRS == 108, na.rm = TRUE)
  )
floor_ceiling_base

```

```

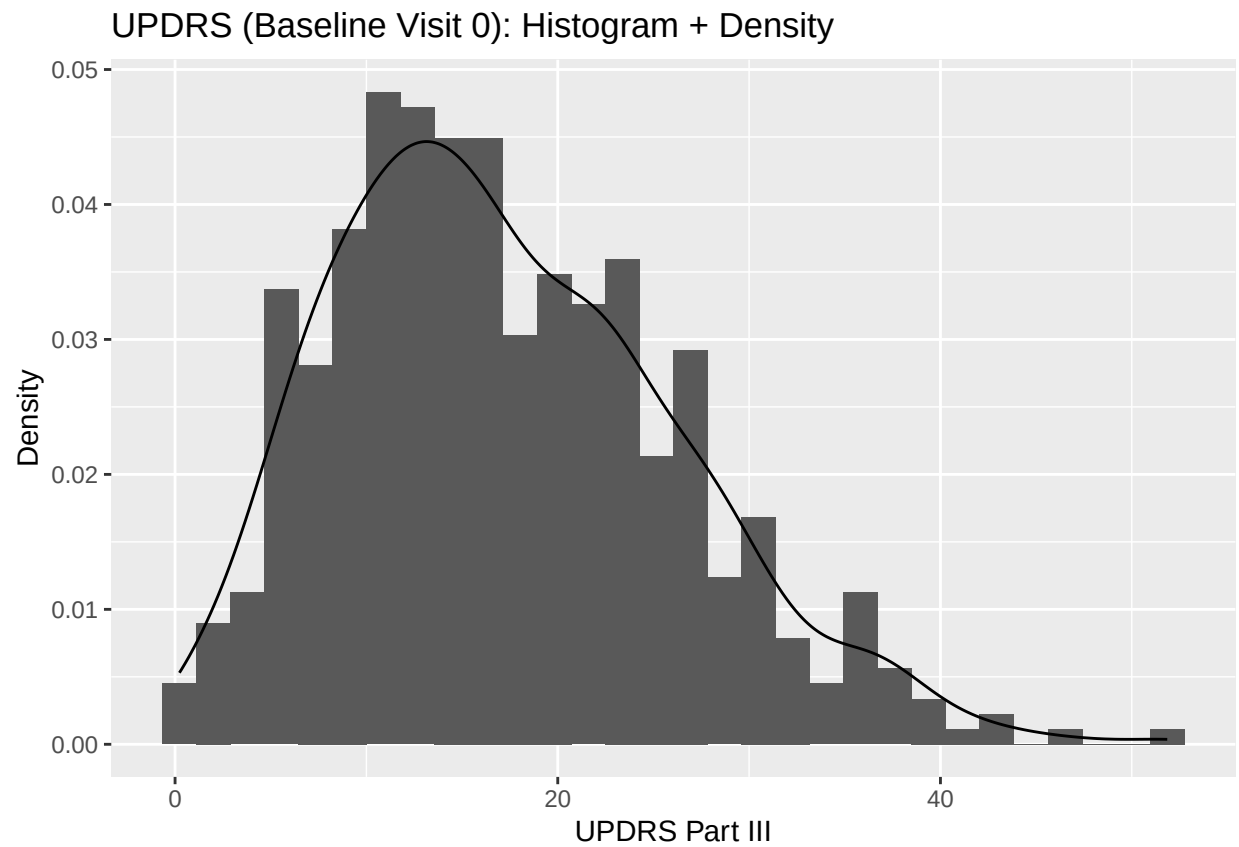
##   floor_n floor_pct ceiling_n ceiling_pct
## 1      0         0         0           0

```

```

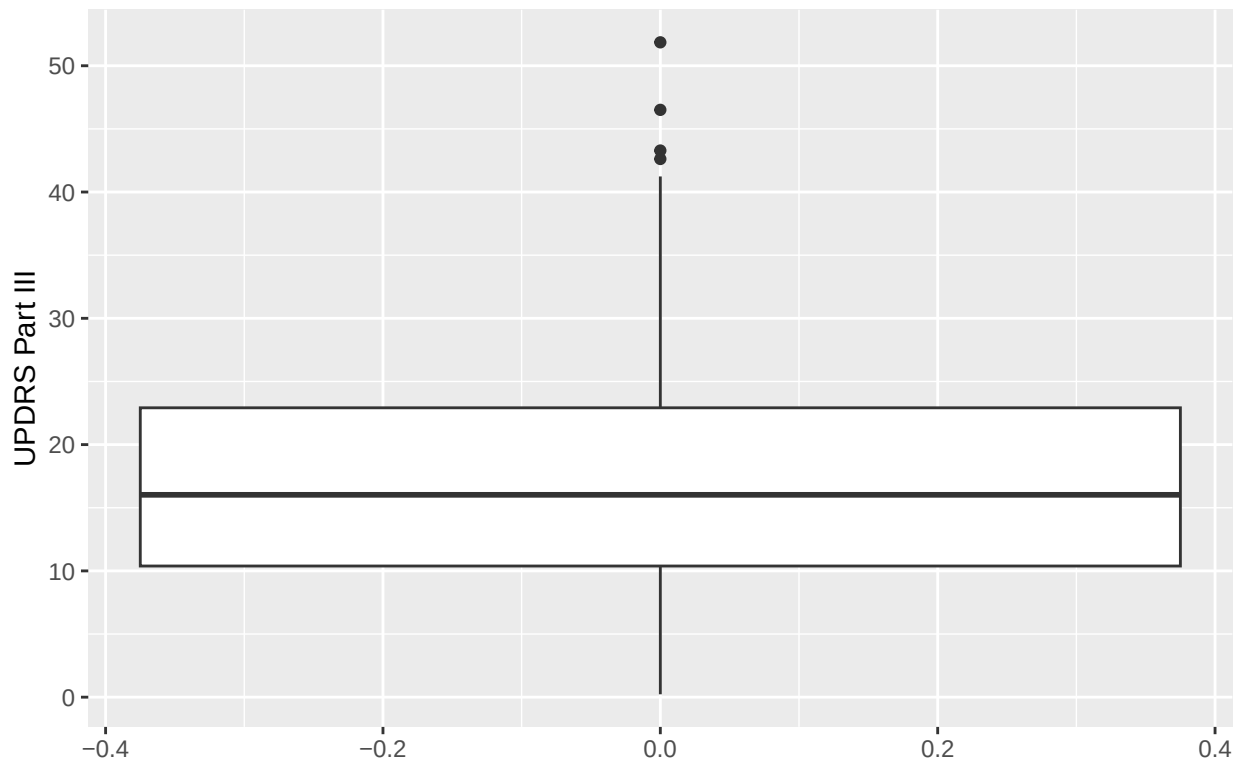
# Baseline plots
ggplot(DF_base, aes(x = UPDRS)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30, na.rm = TRUE) +
  geom_density(na.rm = TRUE) +
  labs(title = "UPDRS (Baseline Visit 0): Histogram + Density", x = "UPDRS Part III", y = "Density")

```



```
ggplot(DF_base, aes(y = UPDRS)) +  
  geom_boxplot(na.rm = TRUE) +  
  labs(title = "UPDRS (Baseline Visit 0): Boxplot", x = "", y = "UPDRS Part III")
```

UPDRS (Baseline Visit 0): Boxplot



Visits

The number of observed UPDRS measurements varied across visits, reflecting attrition and intermittent missingness over time. At baseline (Visit 0), all 500 patients were assessed. At subsequent visits, the number of observations ranged between 303 and 342, corresponding to approximately 61–68% of the original cohort per visit. The lowest number of observations occurred at Visit 2 ($n = 303$), while Visit 3 had the highest follow-up count ($n = 342$). The absence of a strictly decreasing pattern suggests that missingness is not purely monotone dropout; rather, some patients appear to have intermittent missing visits and later return for assessment. Overall, although follow-up is incomplete across time, a substantial proportion of the cohort contributes data at each visit, which supports the feasibility of longitudinal mixed-effects modeling under a Missing at Random assumption.

```
# Table of visit counts
visit_counts <- DF %>%
  count(Visit, name = "n") %>%
  arrange(Visit)
visit_counts
```

```
##   Visit    n
## 1      0 500
## 2      1 326
## 3      2 303
## 4      3 342
## 5      4 327
## 6      5 335
## 7      6 314
## 8      7 315
```

```
# Confirm values 0-7 and flag unexpected levels
DF %>%
  filter(is.na(Visit) | Visit < 0 | Visit > 7) %>%
  distinct(Visit) %>%
  arrange(Visit)
```

```
## [1] Visit
## <0 rows> (or 0-length row.names)
```

Bivariate Descriptive Statistics

With the bivariate exploration, we will focus on:

- How does UPDRS change over time?
- Does that change differ by smoking status?
- Is UPDRS associated with age or sex?
- Are trajectories roughly linear or not?

```
library(GGally)
```

```
## Warning: package 'GGally' was built under R version 4.4.3
```

```
library(lme4)
```

```
## Warning: package 'lme4' was built under R version 4.4.3
```

```
## Loading required package: Matrix
```

```
## Warning: package 'Matrix' was built under R version 4.4.3
```

```
##
```

```
## Attaching package: 'Matrix'
```

```
## The following objects are masked from 'package:tidyr':
```

```
##
```

```
##      expand, pack, unpack
```

```
library(lmerTest)
```

```
## Warning: package 'lmerTest' was built under R version 4.4.3
```

```
##
```

```
## Attaching package: 'lmerTest'
```

```
## The following object is masked from 'package:lme4':
```

```
##
```

```
##      lmer
```

```
## The following object is masked from 'package:stats':
```

```
##
```

```
##      step
```

Mean Structure Over Time

The goal of this section is to understand the average UPDRS trajectory, i.e. the **mean UPDRS by visit**, individual trajectories (with spaghetti plots), mean trajectory by **smoking status**, and mean trajectory by **sex**.

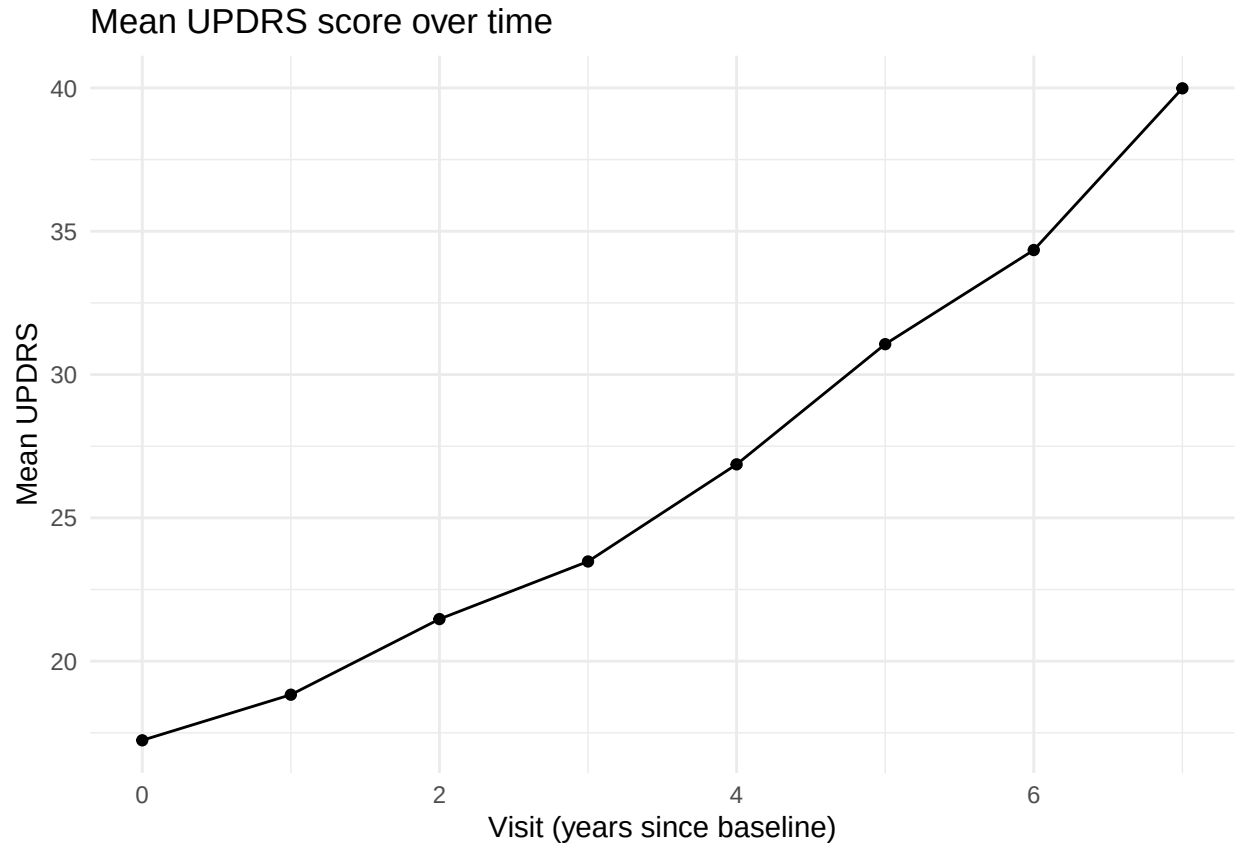
We aim to understand whether the disease progression is roughly linear, whether groups differ, and interaction terms are needed.

```
updrs_m_visit <- DF %>%
  group_by(Visit) %>%
  summarise(
    m_UPDRS = mean(UPDRS, na.rm = TRUE),
    sd_UPDRS = sd(UPDRS, na.rm = TRUE),
    n = n()
  )

updrs_m_visit
```

```
## # A tibble: 8 x 4
##   Visit m_UPDRS sd_UPDRS     n
##   <int>   <dbl>   <dbl> <int>
## 1     0    17.2     8.94   500
## 2     1    18.8    10.9   326
## 3     2    21.5    12.2   303
## 4     3    23.5    13.4   342
## 5     4    26.9    15.4   327
## 6     5    31.1    18.3   335
## 7     6    34.3    17.2   314
## 8     7    40.0    18.6   315
```

```
ggplot(updrs_m_visit, aes(x = Visit, y = m_UPDRS)) +
  geom_line() +
  geom_point() +
  labs(
    title = "Mean UPDRS score over time",
    x = "Visit (years since baseline)",
    y = "Mean UPDRS"
  ) +
  theme_minimal()
```



The mean UPDRS Part III score increases steadily across the follow-up visits, indicating a clear progression of motor symptom severity over time. At baseline (Visit 0), the mean UPDRS score is 17.24 (SD = 8.94). The mean score increases gradually during the first years of follow-up, reaching 21.47 at Visit 2 and 26.87 at Visit 4. In the later visits, the increase becomes more pronounced, with mean scores of 31.06, 34.34, and 39.99 at Visits 5, 6, and 7 respectively.

This monotonic increase in the mean UPDRS score suggests a **worsening of motor symptoms as the disease progresses**. The trajectory appears approximately linear, although the slope seems slightly steeper in the later follow-up years, which may indicate an acceleration of symptom progression over time, i.e. a **slightly non-linear trajectory**.

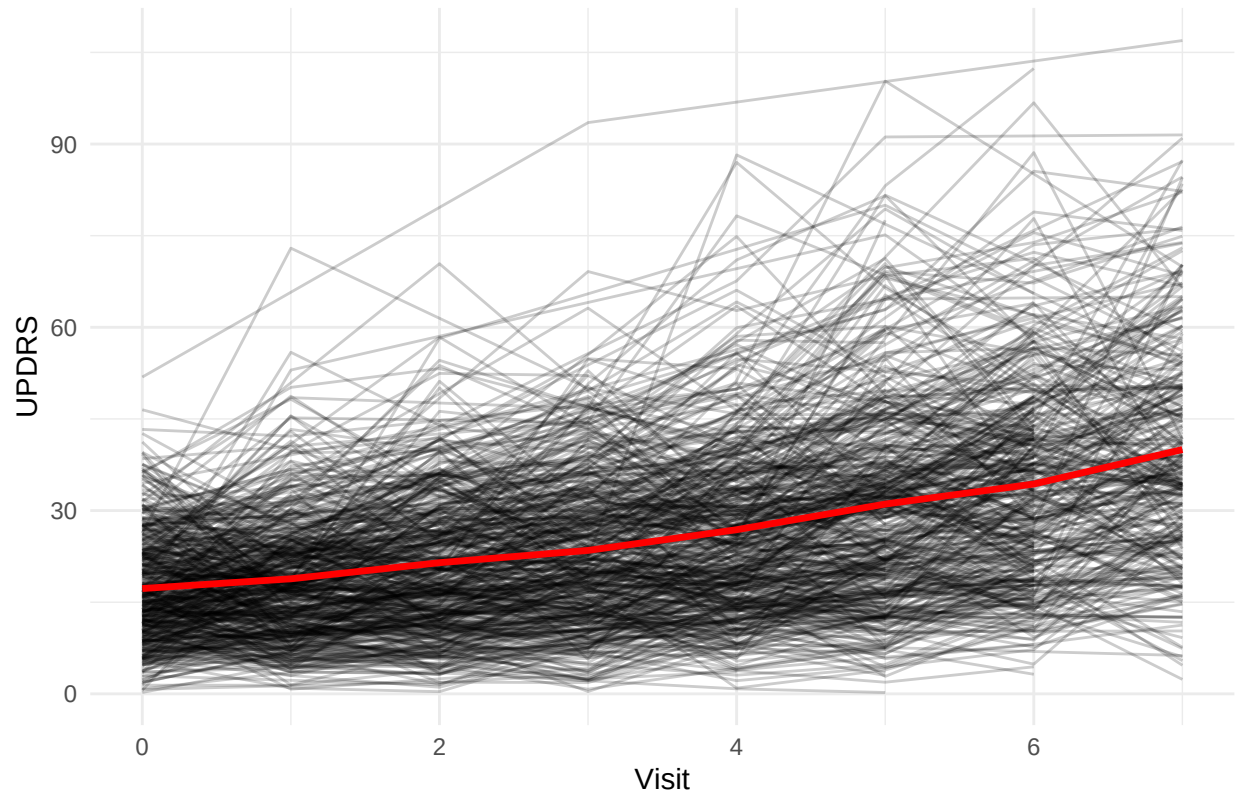
The variability of the UPDRS scores also increases over time. The standard deviation rises from 8.94 at baseline to approximately 18–19 in the later visits, suggesting increasing heterogeneity between patients as the disease progresses.

Finally, the number of observations decreases across visits (from 500 at baseline to around 315 at Visit 7), reflecting **attrition** in the cohort over time. This reduction is typical in long-term longitudinal studies and may arise from dropout, loss to follow-up, or inability to complete assessments as the disease advances.

```
ggplot(DF, aes(x = Visit, y = UPDRS, group = id)) +  
  geom_line(alpha = 0.2) +  
  stat_summary(aes(group = 1),  
    fun = mean,  
    geom = "line",  
    color = "red",  
    linewidth = 1.2) +  
  labs(  
    title = "Individual trajectories of UPDRS",
```

```
x = "Visit",
y = "UPDRS"
) +
theme_minimal()
```

Individual trajectories of UPDRS



Individual trajectories The spaghetti plot displays the longitudinal UPDRS trajectories for all patients over the 7-year follow-up period. Each grey line represents the repeated measurements of a single patient, while the red line corresponds to the average trajectory across all patients.

Overall, the average UPDRS score increases steadily over time, indicating a clear worsening of motor symptoms as the disease progresses. This pattern is consistent with the mean profile plot presented earlier and reflects the expected progression of Parkinson's disease.

However, the individual trajectories reveal substantial heterogeneity between patients. At baseline, patients already exhibit considerable variability in UPDRS scores, with values ranging from near zero to above 40. This variability persists and appears to increase over time, as the spread of the trajectories widens in later visits.

In addition to differences in baseline levels, patients also appear to differ in their rates of progression. While most trajectories show an upward trend, some patients experience relatively slow progression, whereas others show a much steeper increase in UPDRS scores over time. The presence of both vertical shifts and differing slopes suggests that patients not only start at different levels of motor impairment but may also deteriorate at different rates.

Random Intercept

The clear vertical separation between trajectories at baseline suggests that patients differ substantially in their initial level of motor impairment. This indicates that a random intercept for subjects should be included in the model to account for between-patient variability in baseline UPDRS scores.

In practice, this corresponds to including a subject-specific intercept: $(1 \mid \text{id})$. This allows each patient to have their own starting level of UPDRS while sharing the same fixed-effect mean trajectory.

Random Slope for Time

The trajectories also show noticeable differences in the rate of increase over time, with some patients progressing faster than others. This suggests that allowing the effect of time to vary between individuals may improve the model fit.

This would correspond to including a random slope for Visit: $(\text{Visit} \mid \text{id})$. Such a structure allows each patient to have both their own baseline level and their own progression rate.

However, whether random slopes are required will ultimately be assessed through model comparison and diagnostics. We use the spaghetti plot to provide a visual indication that they may be appropriate.

Mean structure

Both the mean trajectory plot and the red line in the spaghetti plot suggest that UPDRS increases approximately linearly over time, although the increase appears slightly steeper in later visits. As a starting point, a linear time effect is therefore a reasonable assumption for the mean structure.

A basic fixed-effects structure would therefore include:

- time (Visit)
- smoking status
- age
- sex

and potentially the interaction between time and smoking status, since the research question focuses on whether smoking influences the disease trajectory.

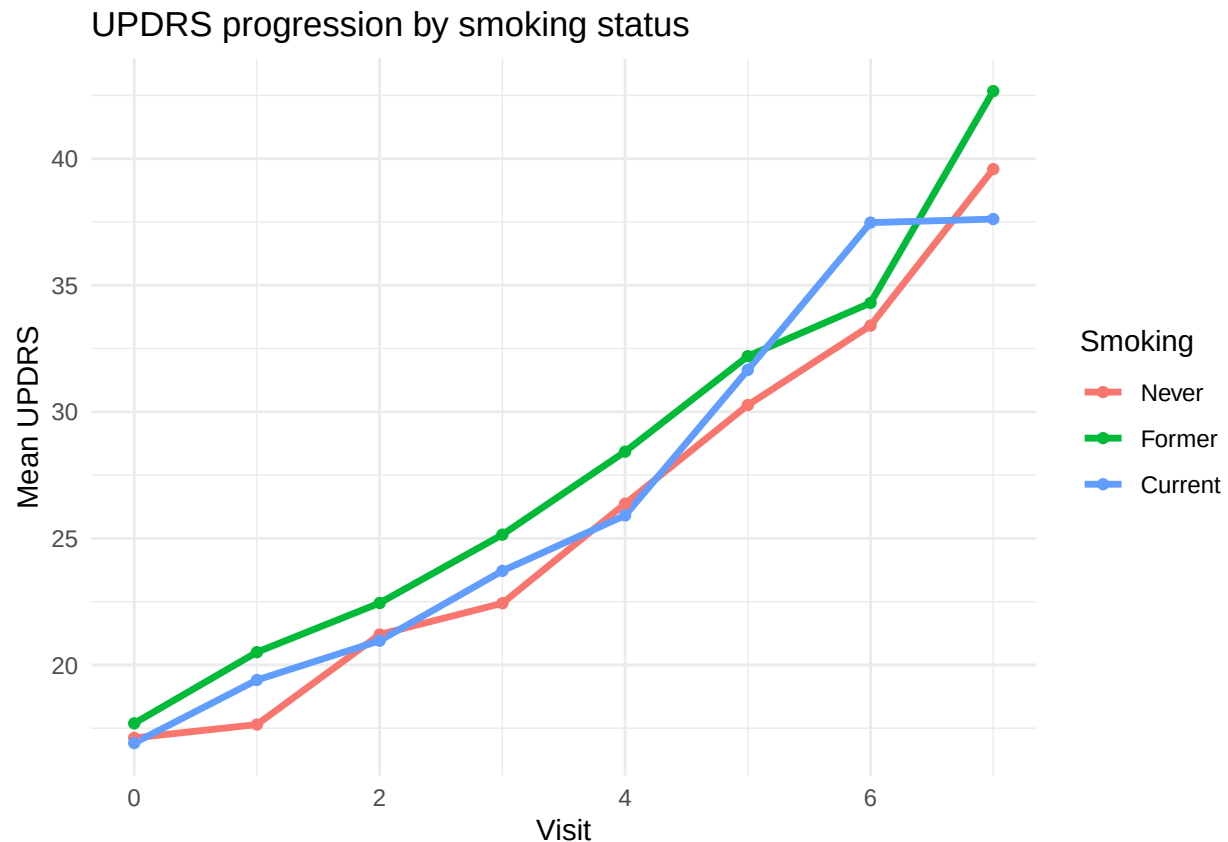
This leads to a mean model of the form: $\text{UPDRS} \sim \text{Visit} + \text{smoking_status} + \text{Age} + \text{sex}$

or, if the effect of smoking is hypothesized to change over time: $\text{UPDRS} \sim \text{Visit} * \text{smoking_status} + \text{Age} + \text{sex}$.

Smoking and Disease Progression

```
ggplot(DF,
  aes(x = Visit, y = UPDRS, color = smoking_status)) +
  stat_summary(fun = mean, geom = "line", size = 1.2) +
  stat_summary(fun = mean, geom = "point") +
  labs(
    title = "UPDRS progression by smoking status",
    x = "Visit",
    y = "Mean UPDRS",
    color = "Smoking"
  ) +
  theme_minimal()
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



The figure displays the mean UPDRS trajectory over time stratified by smoking status (never smokers, former smokers, and current smokers). At baseline, the mean UPDRS scores are relatively similar across the three groups, with values around 17–18. Former smokers appear to have slightly higher baseline scores than the other groups, while never smokers and current smokers show nearly identical initial levels. These small differences suggest that baseline motor symptom severity does not differ substantially by smoking status.

Over time, UPDRS scores increase in all three groups, reflecting the progressive nature of Parkinson's disease. However, the rate of increase appears to differ somewhat between smoking groups. Former smokers tend to exhibit slightly higher UPDRS scores throughout follow-up, and their trajectory appears somewhat steeper in the later visits, reaching the highest mean value at the final visit. Current smokers show a trajectory that increases steadily during the early visits but appears to plateau slightly toward the end of follow-up. Never smokers show a relatively smooth increase over time, with intermediate UPDRS levels compared to the other groups.

The trajectories for the three smoking groups are not perfectly parallel, particularly in the later visits where the differences between groups become more pronounced. This lack of parallelism suggests that the progression of motor symptoms may differ depending on smoking status. In particular, the steeper increase observed among former smokers in the later visits may indicate faster progression compared with the other groups.

Overall, these patterns suggest the possible presence of an interaction between time and smoking status, meaning that the effect of smoking status on UPDRS may change over time. Therefore, it would be reasonable

to consider including a $\text{time} \times \text{smoking_status}$ interaction term in the mixed-effects model to formally test whether disease progression differs between smoking groups.

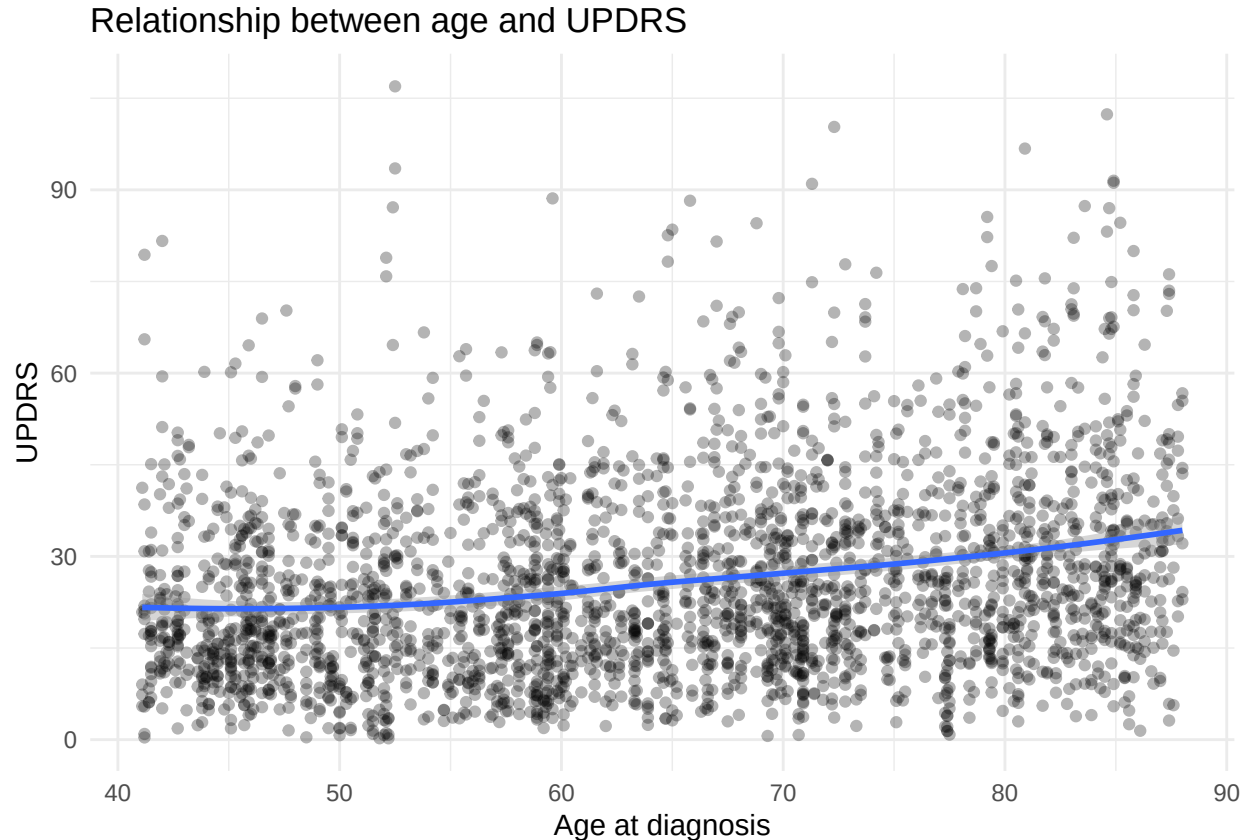
Because the number of observations decreases over time in this cohort, the differences between smoking groups at later visits should be interpreted cautiously. Attrition may influence the observed mean trajectories, particularly in the final follow-up years.

*For model selection, this would mean: $\text{UPDRS} \sim \text{Visit} * \text{smoking_status} + \text{Age} + \text{sex}$, because baseline differences exist (main effect of smoking), slopes appear different (interaction), and trajectories are not fully parallel).*

Relationship between UPDRS and Age

```
ggplot(DF, aes(x = Age, y = UPDRS)) +  
  geom_point(alpha = 0.3) +  
  geom_smooth(method = "loess") +  
  labs(  
    title = "Relationship between age and UPDRS",  
    x = "Age at diagnosis",  
    y = "UPDRS"  
  ) +  
  theme_minimal()
```

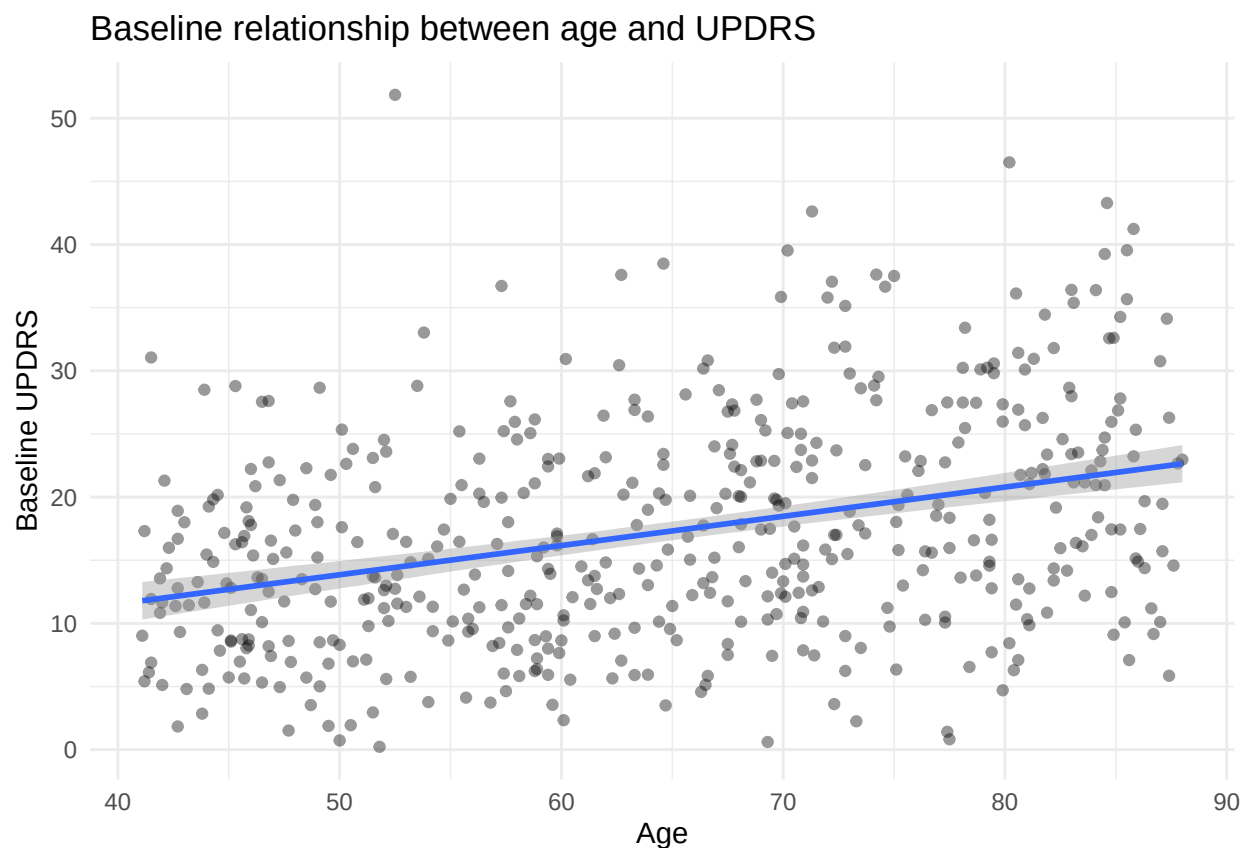
```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
baseline <- DF %>% filter(Visit == 0)

ggplot(baseline, aes(x = Age, y = UPDRS)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm") +
  labs(
    title = "Baseline relationship between age and UPDRS",
    x = "Age",
    y = "Baseline UPDRS"
  ) +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Overall relationship

The first plot shows the relationship between age at diagnosis and UPDRS scores across all visits. The scatter of points reveals a wide variability in UPDRS values for all age groups, indicating substantial heterogeneity in motor symptom severity among patients. Nevertheless, the fitted regression line suggests a positive association between age and UPDRS: older patients tend to have higher UPDRS scores on average.

However, the association appears relatively modest. The slope of the fitted line is shallow, and the large dispersion of observations around the line indicates that age explains only a limited portion of the variability in motor symptom severity. This is expected in longitudinal clinical data, where disease progression and individual variability play important roles.

Another important aspect of this plot is that it combines all repeated measurements from all visits, meaning that the same patients contribute multiple observations. Because UPDRS scores increase over time, later measurements are generally higher. As a result, part of the observed relationship between age and UPDRS may reflect disease progression over time rather than purely baseline age effects.

Overall, the plot suggests that older patients tend to exhibit somewhat higher UPDRS scores, but age alone does not strongly determine symptom severity.

Baseline relationship

The second plot focuses exclusively on baseline measurements (Visit = 0), providing a clearer view of the relationship between age and motor symptom severity at the time of inclusion in the study.

At baseline, a positive association between age and UPDRS is also observed. Older patients tend to have slightly higher baseline UPDRS scores compared with younger patients. The fitted regression line again shows a mild upward trend, indicating that motor symptom severity at diagnosis tends to increase with age.

Despite this trend, the variability around the regression line remains substantial. Patients of similar age can have very different UPDRS scores, suggesting that factors other than age contribute importantly to baseline motor impairment.

Compared with the overall plot, the baseline relationship appears slightly clearer, as it is not influenced by the longitudinal progression of the disease or repeated measurements. This confirms that age at diagnosis is modestly associated with baseline motor symptom severity.

Together, these plots suggest that age has a positive but moderate association with UPDRS scores. The relationship appears approximately linear, with no clear indication of strong non-linearity. Therefore, age can reasonably be included as a linear covariate in the mixed-effects model

In the context of the longitudinal model, age will likely act as a baseline predictor of motor symptom severity, adjusting for differences between patients at study entry while the progression of symptoms over time is captured by the time variable.

Now the question arises: Do older patients deteriorate faster, or do they simply start at a worse level? as in $UPDRS = \beta_0 + \beta_1 * Visit$

```
library(dplyr)
library(broom)
```

```
## Warning: package 'broom' was built under R version 4.4.3
```

```
patient_slopes <- DF %>%
  group_by(id) %>%
  do(tidy(lm(UPDRS ~ Visit, data = .))) %>%
  filter(term == "Visit") %>%
  select(id, slope = estimate)

#Add patient age

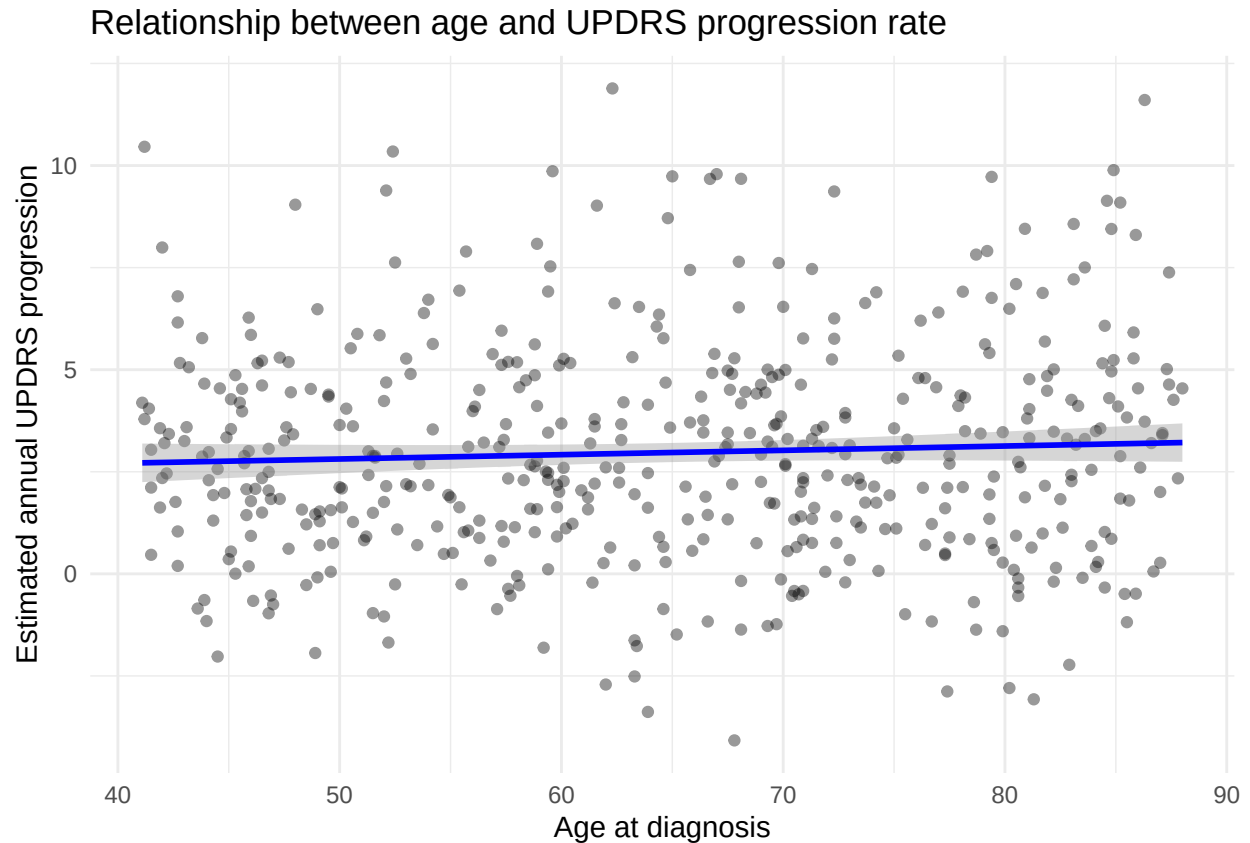
patient_age <- DF %>%
  filter(Visit == 0) %>%
  select(id, Age)

slope_data <- left_join(patient_slopes, patient_age, by = "id")

ggplot(slope_data, aes(x = Age, y = slope)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = TRUE, color = "blue") +
```

```
labs(
  title = "Relationship between age and UPDRS progression rate",
  x = "Age at diagnosis",
  y = "Estimated annual UPDRS progression"
) +
theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



The relationship between age at diagnosis and the estimated annual progression rate of UPDRS was explored by calculating patient-specific slopes from simple linear regressions of UPDRS on time. The resulting scatterplot shows substantial variability in progression rates across patients, indicating heterogeneous disease trajectories. Although the fitted trend line suggests a very slight positive association between age and progression rate, the relationship appears weak and the variability around the line is large. This suggests that age may be modestly associated with baseline motor symptom severity but does not strongly influence the rate of disease progression. Consequently, age is likely to act primarily as a baseline covariate in the longitudinal model rather than interacting strongly with time.